APMA 4302 Final Project

Computational Study of Pulse Propagation in Homogeneous, Layered, and Randomly Layered Media

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Course: APMA 4302

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1 Introduction

The goal of this project is to study numerical wave propagation in different media. More specifically, I want to understand how pulse propagation changes in homogeneous, layered, and randomly layered media, with a focus on transmission, reflection, and waveform distortion.

The computational goal is to use PETSc as the main implementation tool and to build a forward solver for a one-dimensional acoustic pressure equation. I begin with the homogeneous case as a baseline, and then compare it with deterministic layered and randomly layered media.

The project has four main parts:

1. define a one-dimensional acoustic pressure model,
2. discretize it with an explicit finite difference method,
3. implement the solver in PETSc/`petsc4py`,
4. compare the numerical results across the three media.

This report presents the model, the numerical method, and the main propagation results for homogeneous, deterministic layered, and randomly layered media.

2 Mathematical Model

I use a one-dimensional acoustic pressure formulation. The unknown is the acoustic pressure $p(x,t)$, and the governing equation is

$$p_{tt}(x,t) = c(x)^2 p_{xx}(x,t) + s(x,t), \qquad x \in [0,1], \quad t \in [0,T]. \tag{1}$$

Here:

- $p(x,t)$ denotes the acoustic pressure,
- $c(x)$ is the wave speed,
- $s(x,t)$ is the source term.

The initial conditions are chosen to be

$$p(x,0) = 0, \qquad p_t(x,0) = 0. \tag{2}$$

This means that there is no initial disturbance in the medium, and the wave is generated entirely by the source.

For the first implementation, I use simple Dirichlet boundary conditions:

$$p(0,t) = 0, \qquad p(1,t) = 0. \tag{3}$$

These are not the most physically realistic boundary conditions for wave propagation, since they can introduce reflections from the boundaries, but they provide a simple and transparent starting point for the first version of the solver.

The source is modeled as a localized pulse. In the first implementation, I place the source at a single grid point and use a Gaussian time profile:

$$s_i^n = \begin{cases} A \exp\left(-\dfrac{(t^n - t_0)^2}{\sigma^2}\right), & i = i_s, \\ 0, & i \neq i_s. \end{cases} \tag{4}$$

Here A is the amplitude, t_0 is the center time of the pulse, σ controls the pulse width, and i_s is the source grid point.

In the homogeneous-medium case considered in this first part,

$$c(x) = c_0, \tag{5}$$

where c_0 is constant.

3 Numerical Method

3.1 Grid and notation

The spatial interval $[0,1]$ is divided into N_x subintervals, with grid points

$$x_i = i\Delta x, \qquad i = 0, 1, \ldots, N_x, \tag{6}$$

where

$$\Delta x = \frac{1}{N_x}. \tag{7}$$

Similarly, the time interval $[0,T]$ is divided into time steps

$$t^n = n\Delta t, \qquad n = 0, 1, \ldots, N_t. \tag{8}$$

The discrete approximation to the pressure is denoted by

$$p_i^n \approx p(x_i, t^n). \tag{9}$$

3.2 Finite difference discretization

I use centered finite differences in both time and space. The second derivative in time is approximated by

$$p_{tt}(x_i, t^n) \approx \frac{p_i^{n+1} - 2p_i^n + p_i^{n-1}}{\Delta t^2}, \tag{10}$$

and the second derivative in space is approximated by

$$p_{xx}(x_i, t^n) \approx \frac{p_{i+1}^n - 2p_i^n + p_{i-1}^n}{\Delta x^2}. \tag{11}$$

Substituting these into the PDE gives

$$\frac{p_i^{n+1} - 2p_i^n + p_i^{n-1}}{\Delta t^2} = c_i^2 \frac{p_{i+1}^n - 2p_i^n + p_{i-1}^n}{\Delta x^2} + s_i^n. \tag{12}$$

Solving for p_i^{n+1}, the explicit update formula becomes

$$p_i^{n+1} = 2p_i^n - p_i^{n-1} + \Delta t^2 \left(c_i^2 \frac{p_{i+1}^n - 2p_i^n + p_{i-1}^n}{\Delta x^2} + s_i^n \right). \tag{13}$$

This update is applied at all interior grid points $i = 1, \ldots, N_x - 1$. The boundary values are set directly from the boundary conditions:

$$p_0^n = 0, \qquad p_{N_x}^n = 0. \tag{14}$$

3.3 Initialization of the first time step

Because the explicit update formula requires both p^n and p^{n-1}, the first time level p^1 must be constructed separately. I do this using a Taylor expansion in time:

$$p_i^1 = p_i^0 + \Delta t (p_t)_i^0 + \frac{\Delta t^2}{2} \left(c_i^2 \frac{p_{i+1}^0 - 2p_i^0 + p_{i-1}^0}{\Delta x^2} + s_i^0 \right). \tag{15}$$

Since the initial conditions are $p_i^0 = 0$ and $(p_t)_i^0 = 0$, this simplifies substantially. In particular, because $p^0 \equiv 0$, the discrete Laplacian term is initially zero, so

$$p_i^1 = \frac{\Delta t^2}{2} s_i^0. \tag{16}$$

3.4 Stability and CFL condition

Since the method is explicit, the time step must satisfy a CFL-type stability condition. In one dimension, a standard condition is

$$\frac{c_{\max}\Delta t}{\Delta x} \leq 1, \tag{17}$$

where $c_{\max} = \max_x c(x)$.

To remain on the safe side, I choose

$$\Delta t = 0.5\frac{\Delta x}{c_{\max}}. \tag{18}$$

This conservative choice helps keep the numerical scheme stable. In practice, the value of $c_{\max}$ is recomputed separately for each medium. As a result, the homogeneous, deterministic layered, and randomly layered simulations do not all use exactly the same Δt; instead, each run uses a time step based on the maximum wave speed in that specific medium.

3.5 Implementation

The code is implemented in PETSc through `petsc4py`. The spatial operator is assembled as a PETSc sparse matrix, and the second spatial derivative is computed by matrix-vector multiplication at each time step. The main solution vectors are `p_prev`, `p_curr`, and `p_next`, which store the solution at three consecutive time levels.

At each step, the code applies the matrix to the current solution, adds the source term, updates the solution with the explicit second-order scheme, enforces the boundary conditions, and then shifts the vectors forward in time.

4 Homogeneous Medium

4.1 Computational setup

For the first completed baseline experiment, I use the following parameters:

- domain length: $L = 1$,
- number of spatial intervals: $N_x = 400$,
- homogeneous wave speed: $c_0 = 1$,
- source location: $x_s = 0.2$,

- receiver location: $x_r = 0.8$,
- total simulation time: $T = 1.2$,
- source amplitude: $A = 1$,
- source center time: $t_0 = 0.05$,
- source width: $\sigma = 0.01$.

The time step is chosen from the CFL condition, using

$$\Delta t = 0.5 \frac{\Delta x}{c_{\max}}.$$

4.2 Summary of observations

The homogeneous simulation produces a clear forward-propagating pulse and a meaningful receiver signal, which shows that the PETSc-based solver is working at a basic level.

The snapshots show that the wave moves to the right from the source location near $x = 0.2$, which is consistent with finite-speed propagation. The receiver signal also shows a clear main arrival. Since the source is near $x_s = 0.2$, the receiver is near $x_r = 0.8$, and $c_0 = 1$, a simple estimate gives an arrival time of about $0.05 + 0.6 = 0.65$, which agrees with the numerical result.

At the same time, the waveform is broader than an ideal narrow pulse. In the current implementation, the source is injected through a time-dependent forcing term over multiple time steps, so the solution develops a wider, plateau-like shape instead of a sharply localized Gaussian pulse. The later part of the receiver signal is also affected by boundary reflections because of the Dirichlet boundary conditions.

Overall, this homogeneous case provides a useful baseline for comparison with the deterministic layered and randomly layered media.

4.3 Figures

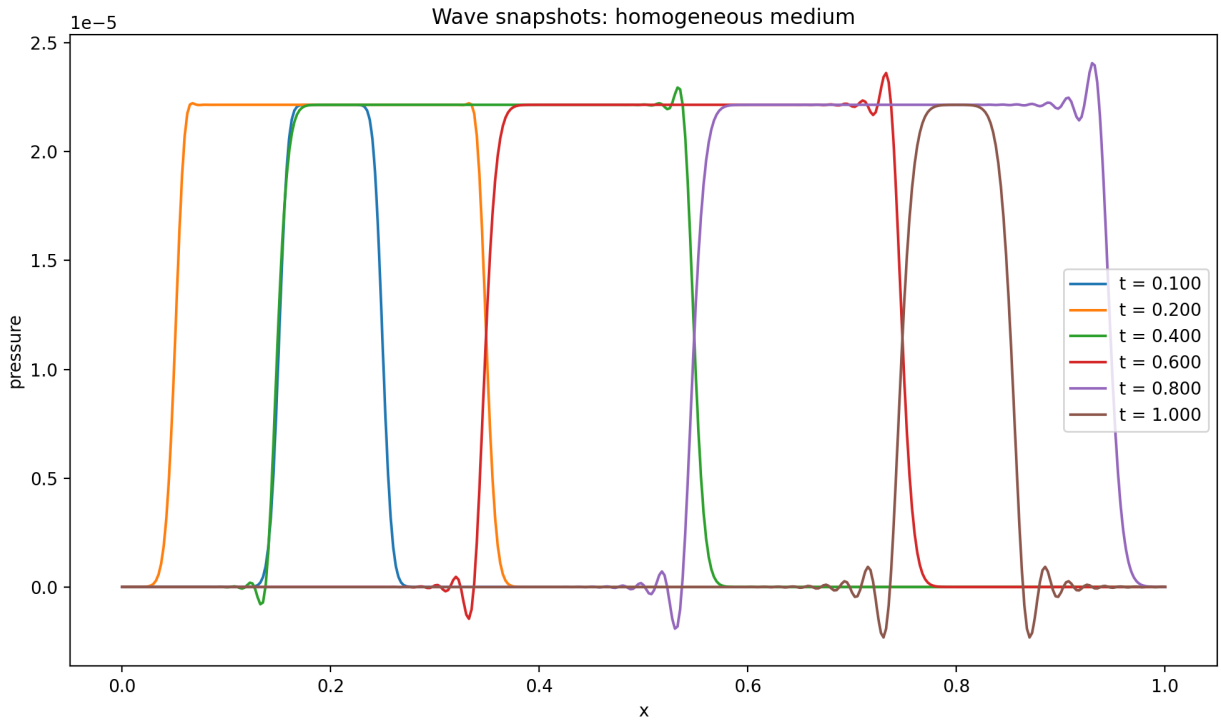

Figure 1: wave snapshots

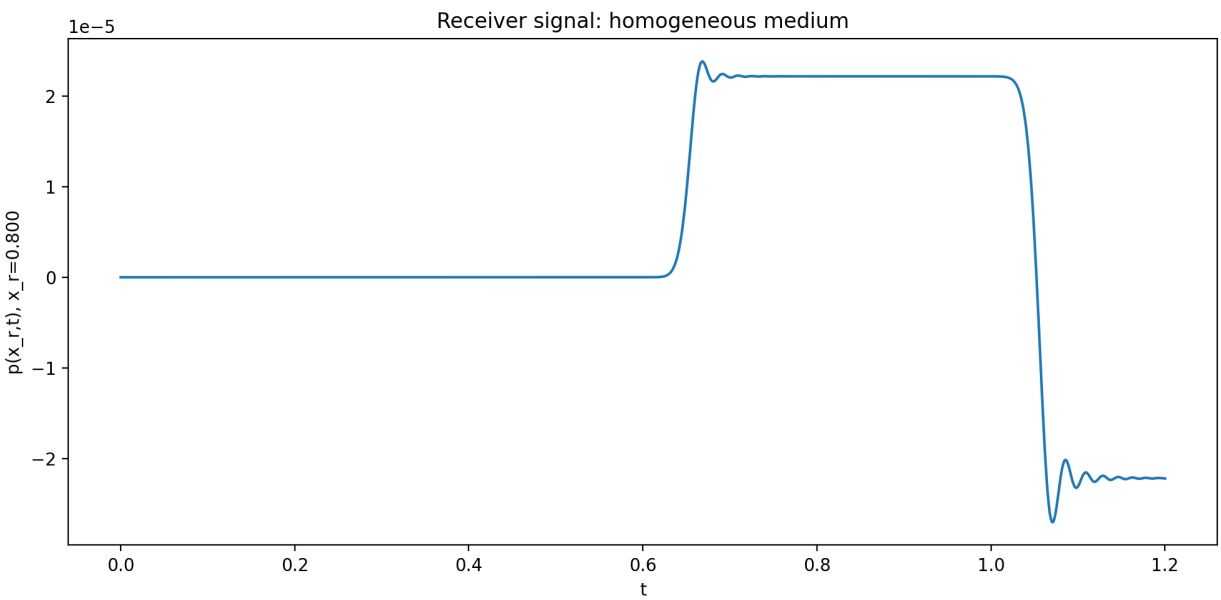

Figure 2: receiver signal

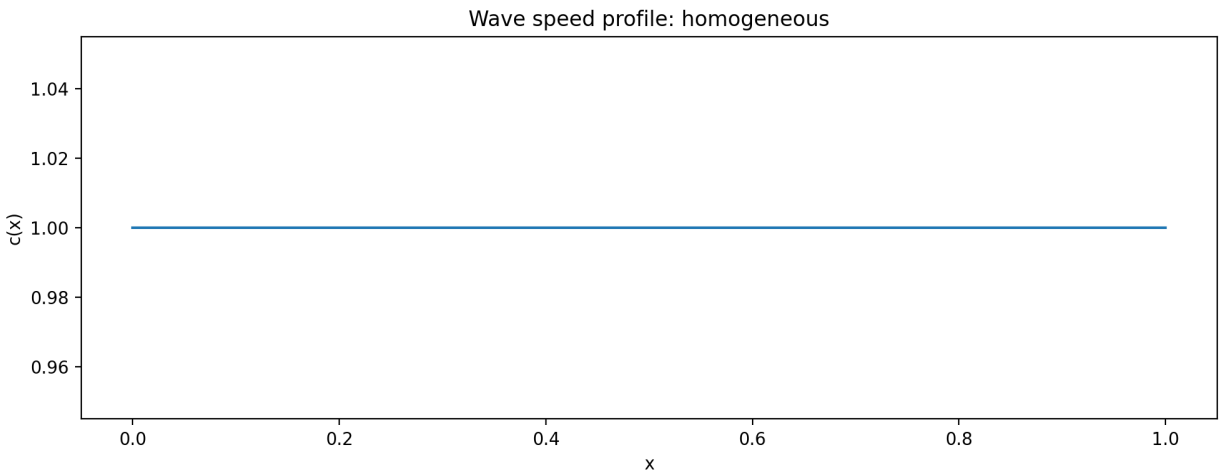

Figure 3: wave speed profile

4.4 Interpretation

Overall, the homogeneous-medium run serves as a successful baseline case. It demonstrates that:

- the discrete model is producing nontrivial wave motion,
- the source injection is working,
- the PETSc-based matrix-vector implementation is functioning correctly,
- the explicit time-stepping scheme remains stable under the chosen CFL condition,
- the receiver signal captures a physically meaningful arrival event.

This provides a solid foundation for the next stages of the project, where the same code structure will be extended to deterministic layered media and randomly layered media.

5 Deterministic Layered Medium

5.1 Computational setup

To extend the homogeneous baseline, I next considered a deterministic layered medium. The wave speed was chosen to be piecewise constant:

$$c(x) = \begin{cases} 1.0, & 0 \leq x < 0.35, \\ 0.7, & 0.35 \leq x < 0.65, \\ 1.2, & 0.65 \leq x \leq 1. \end{cases} \tag{19}$$

All other numerical parameters were kept the same as in the homogeneous case, including the domain length, source location, receiver location, total simulation time, and source pulse parameters. This makes it possible to compare the effect of the medium structure directly.

5.2 Summary of observations

The deterministic layered simulation is more complicated than the homogeneous baseline. In the snapshots, the pulse no longer moves as a simple translated waveform. Its shape changes as it crosses the interfaces between layers with different wave speed.

Visible distortions appear near the interfaces around $x = 0.35$ and $x = 0.65$, which is consistent with partial transmission and reflection at jumps in the medium. Compared with the homogeneous case, the waveform becomes less uniform and develops more oscillatory structure.

The receiver signal also becomes more complicated. A main arrival is still clearly visible, but the later part of the signal is more irregular, which suggests that the layered medium is modifying the transmitted pulse and adding reflected components.

5.3 Figures

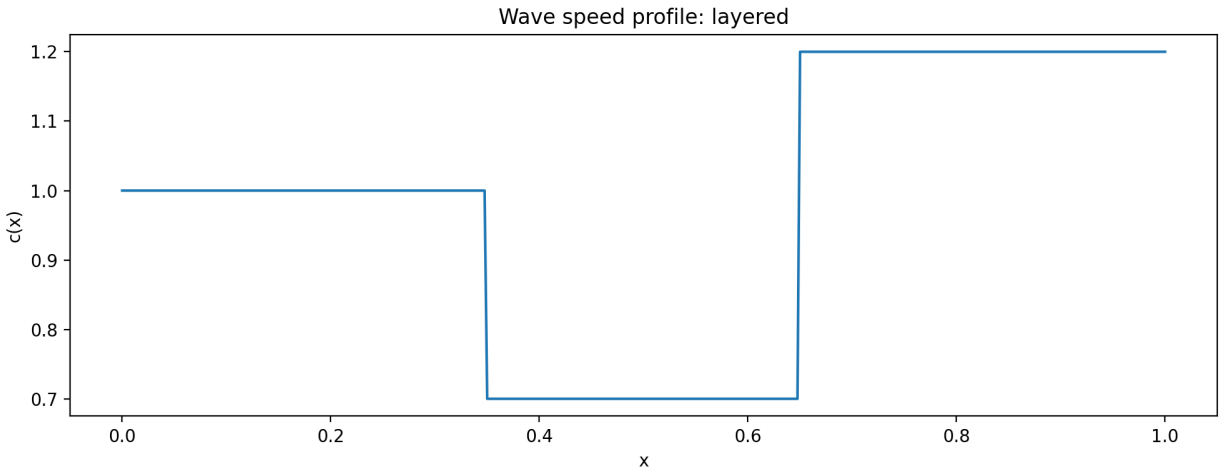

Figure 4: Wave speed profile in the deterministic layered medium. The medium consists of three regions with piecewise constant wave speed.

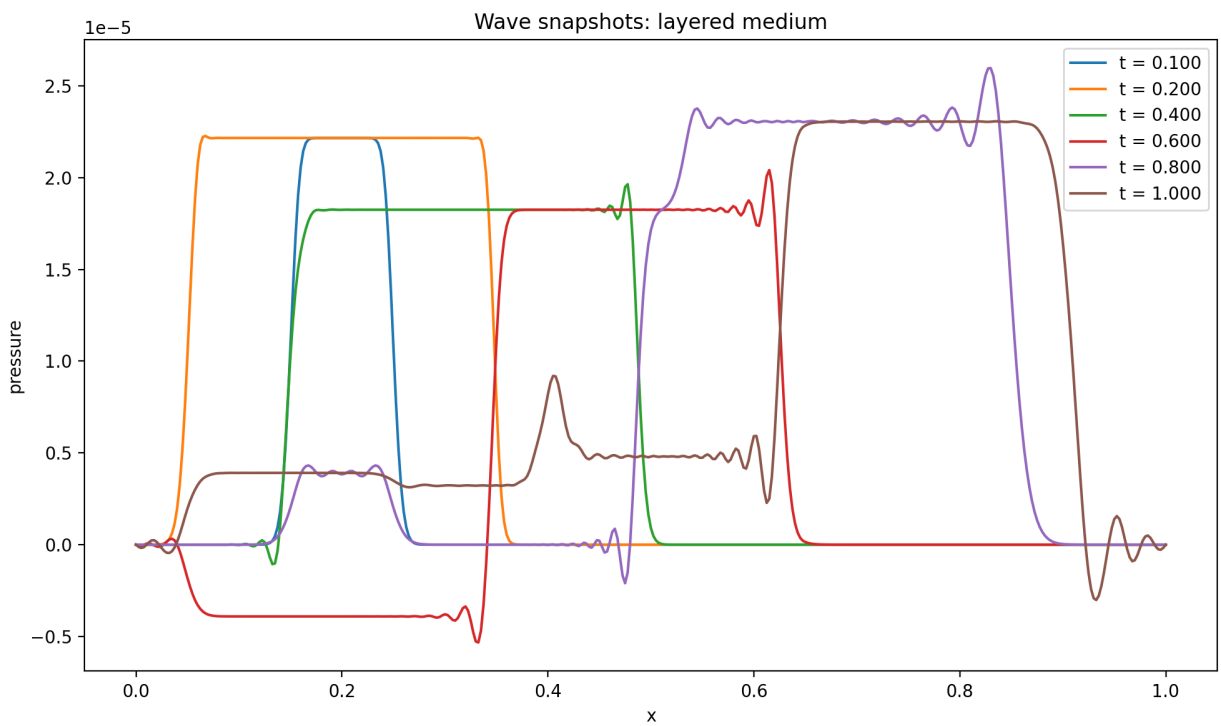

Figure 5: Wave snapshots in the deterministic layered medium at several times. The wave form changes as it crosses the interfaces, and additional oscillatory structure appears.

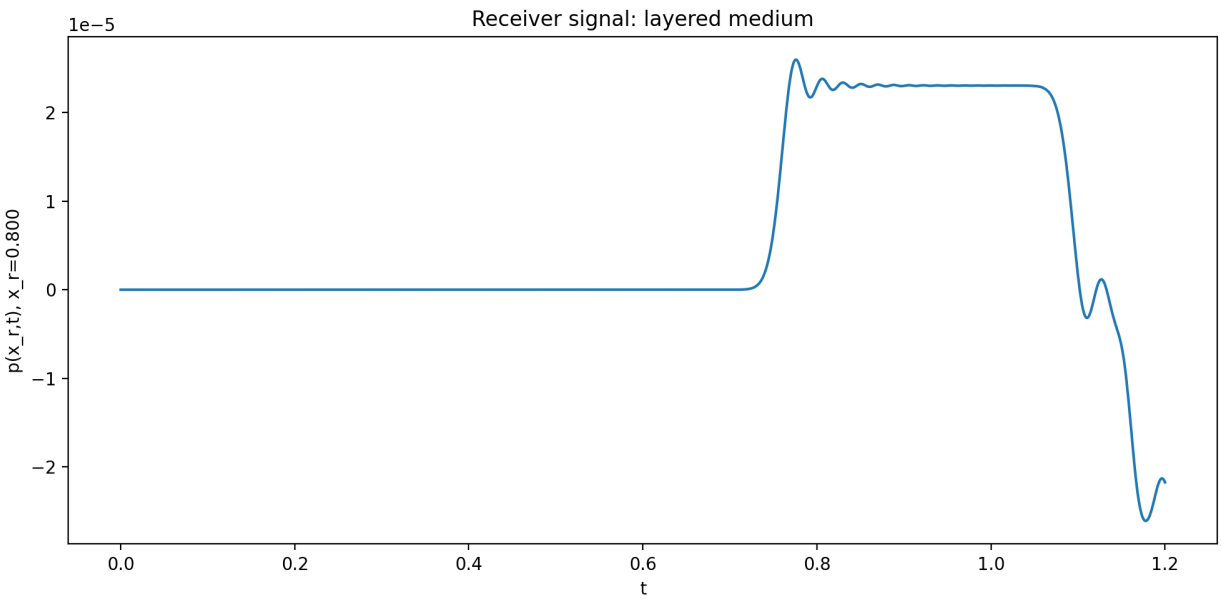

Figure 6: Receiver signal recorded at xr = 0.8 in the deterministic layered medium. Compared with the homogeneous case, the signal shows stronger distortion and more complicated late-time structure.

5.4 Interpretation

Overall, the deterministic layered case confirms that the numerical solver responds meaningfully to changes in medium structure. Compared with the homogeneous baseline, the layered medium produces more complicated propagation because changes in wave speed lead to partial reflection and transmission at the interfaces. This is an important intermediate step before moving on to the randomly layered case.

6 Randomly Layered Medium

6.1 Computational setup

After the deterministic layered case, I next considered a randomly layered medium. The domain was divided into many small layers, and each layer was assigned a random piecewise constant wave speed. In the present implementation, the wave speed varies approximately between 0.7 and 1.3, producing a heterogeneous medium with many local jumps in material properties.

All other numerical parameters were kept the same as in the previous cases, including the domain length, source location, receiver location, total simulation time, and source pulse parameters. This allows the effect of random layering to be compared directly with the homogeneous and deterministic layered cases.

6.2 Summary of observations

The randomly layered simulation produces the most complicated propagation pattern among the three cases. In the snapshots, the pulse no longer looks like a simple translated waveform. Instead, it develops stronger local oscillations and a more irregular structure across the domain.

Compared with the deterministic layered case, the wave now interacts with many more interfaces, so its shape is repeatedly modified during propagation. The pulse becomes more fragmented, and the solution shows more secondary oscillations and local distortions.

The receiver signal is also more irregular than in the previous two cases. A main arrival is still visible, but the signal shows stronger fluctuations and a more complicated late-time structure, which is consistent with repeated scattering in the random medium.

6.3 Figures

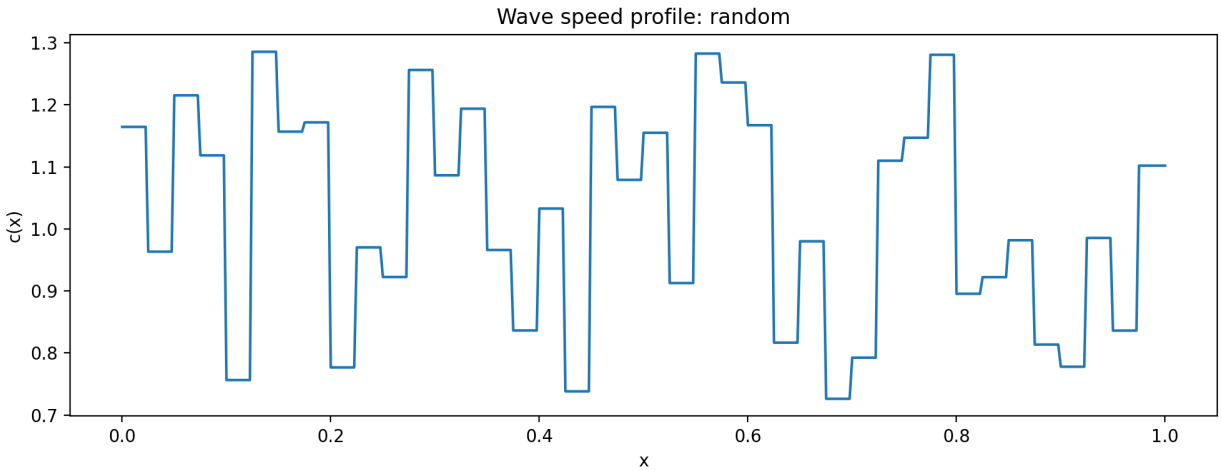

Figure 7: Wave speed profile in the randomly layered medium. The wave speed is piecewise constant but varies randomly from layer to layer.

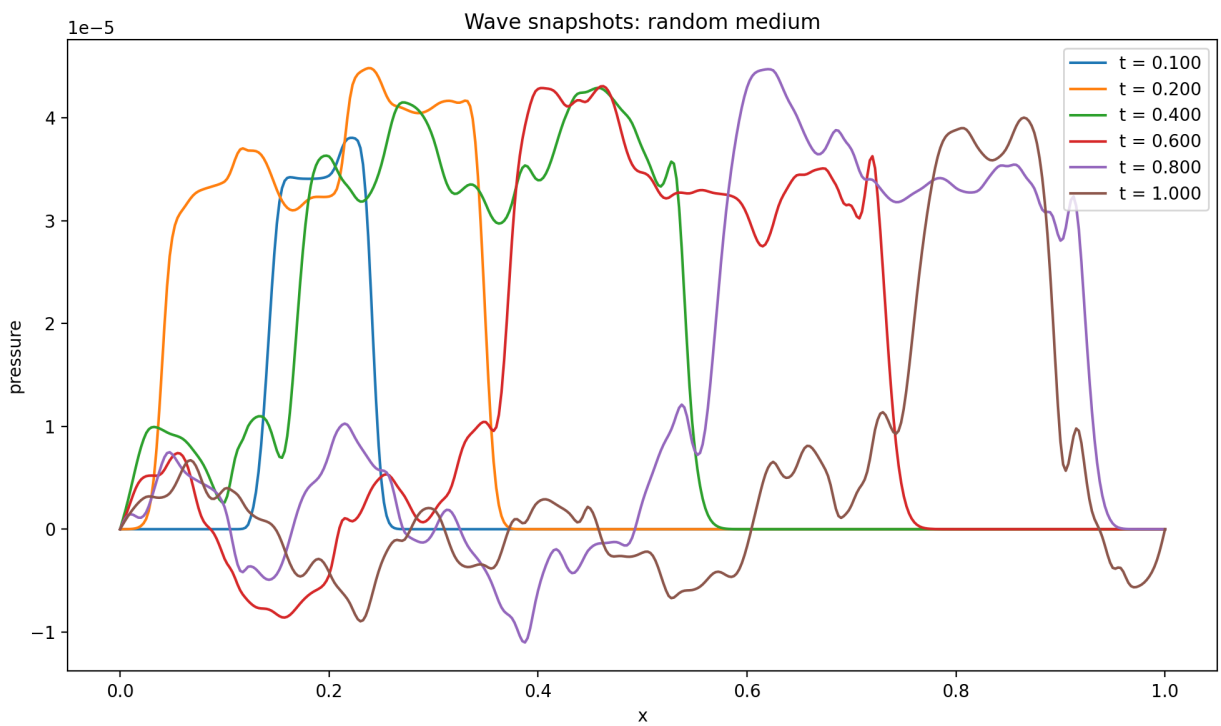

Figure 8: Wave snapshots in the randomly layered medium at several times. Compared with the homogeneous and deterministic layered cases, the waveform develops stronger local oscillations and more irregular structure.

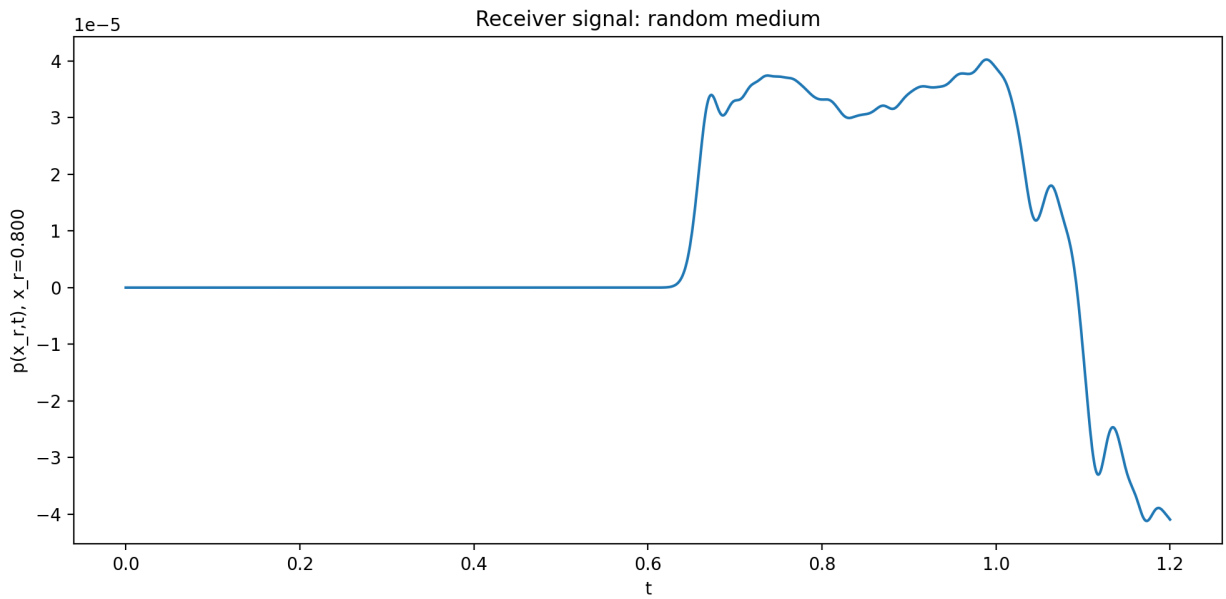

Figure 9: Receiver signal recorded at $x_r = 0.8$ in the randomly layered medium. The signal is more irregular and displays stronger fluctuation than in the previous cases.

6.4 Interpretation

Overall, the randomly layered case shows that the solver is sensitive to fine-scale heterogeneity in the medium. Compared with the homogeneous and deterministic layered cases, the random medium produces more complicated propagation because the wave interacts with many local interfaces rather than only a few large jumps.

The larger amplitude observed in the randomly layered receiver signal should be interpreted cautiously. In the present report, the random medium is represented by a single realization, so the increased amplitude may reflect realization-specific constructive interference or local energy concentration rather than a general statistical conclusion. A more systematic study would require multiple realizations and comparison of ensemble behavior.

7 Grid Sensitivity Check

7.1 Setup

To examine how the numerical results depend on spatial resolution, I repeated the homogeneous-medium simulation using three grid sizes:

$$N_x = 200, \qquad N_x = 400, \qquad N_x = 800.$$

For each run, the time step was recomputed from the CFL condition

$$\Delta t = 0.5 \frac{\Delta x}{c_{\max}},$$

so that the stability requirement remained consistent across resolutions.

The comparison was based on the receiver signal at $x_r = 0.8$, since this provides a direct observable quantity that can be compared across different grids.

7.2 Observations

The three receiver signals show the same main qualitative propagation behavior. In particular, the principal arrival occurs at approximately the same time in all three runs, which is consistent with the expected propagation speed in the homogeneous medium.

At the same time, the amplitudes are not yet fully aligned across grid resolutions. This suggests that the numerical solution still has noticeable grid dependence in its detailed waveform, even though the main arrival structure is already captured consistently. One likely reason is that the current source is injected at a single grid point, so its discrete representation remains mesh-dependent. Therefore, the present figure should be interpreted as a basic grid sensitivity check rather than a strong convergence demonstration.

7.3 Figure

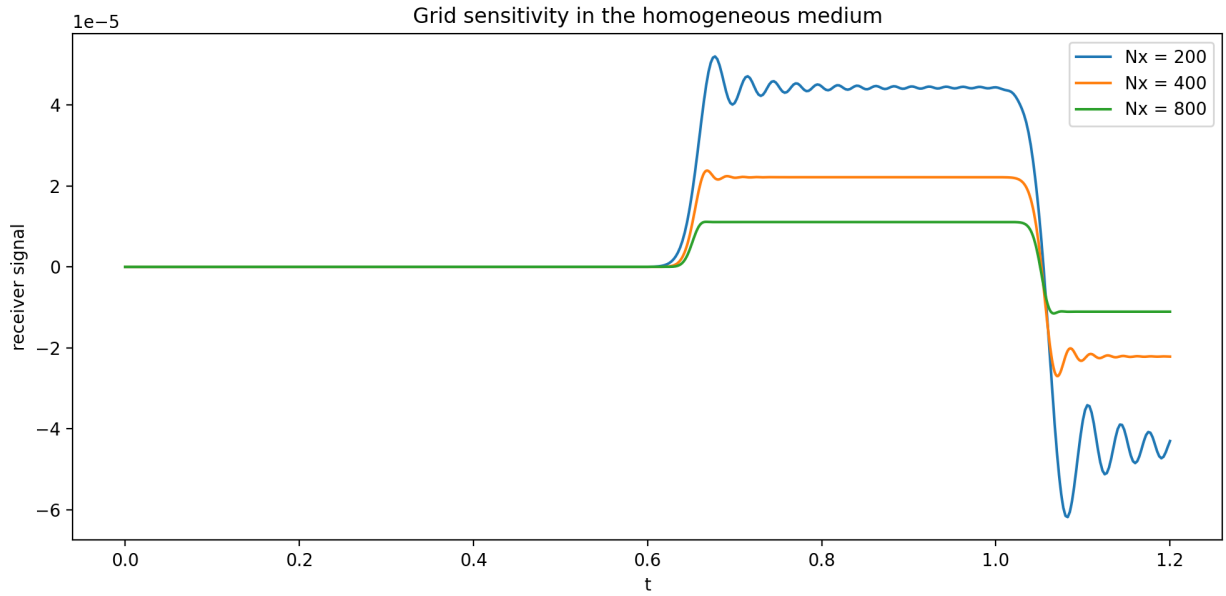

Figure 10: Receiver signal comparison for three grid resolutions in the homogeneous medium. The main arrival time is consistent across grids, although the detailed waveform and amplitude remain resolution-dependent

7.4 Interpretation

This comparison shows that the solver reproduces the same main propagation timing across multiple grids, which supports the qualitative reliability of the homogeneous-medium result. However, because the waveform amplitude still changes noticeably with resolution, this test is better viewed as a consistency check than as a complete convergence verification. A more rigorous convergence study would require either an exact reference solution or a quantitative error analysis between grid levels.

8 Conclusion

In this project, I developed a PETSc-based finite difference solver for a one-dimensional acoustic pressure equation and used it to study pulse propagation in three types of media: homogeneous, deterministic layered, and randomly layered media.

The homogeneous case provided a baseline forward-propagation result and verified that the numerical implementation was working correctly. The deterministic layered case showed that wave-speed jumps modify the waveform through reflection and transmission at interfaces. The randomly layered case produced the most complicated propagation pattern, with

stronger local oscillations and more irregular receiver signals, indicating the cumulative effect of repeated scattering.

Overall, these results show that the solver responds meaningfully to changes in medium structure and provides a useful numerical framework for studying wave propagation in heterogeneous media. A natural direction for future work would be to extend the current propagation study to a basic time-reversal experiment.